

Optimising Weak Reference Processing in OpenJDK's ZGC

Master's Thesis in Computer Science

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Agenda

- 1 Background
- 2 Motivation
- 3 Design & Implementation
- 4 Evaluation
- 5 Results
- 6 Conclusions



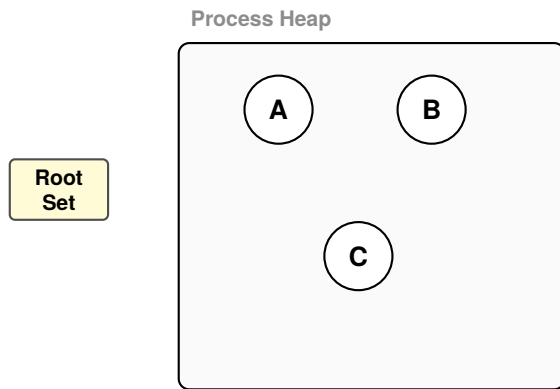
Background

Java and the JVM

- ▶ One of the most used programming languages in the world
- ▶ Write once, compile once – runs on *any* machine via the **JVM**
- ▶ Objects allocated with `new`; no manual memory management
- ▶ **HotSpot JVM** is the reference implementation (Oracle / OpenJDK)
- ▶ Ships with multiple garbage collectors: G1, **ZGC**, ...
- ▶ The user selects the GC at start-up with e.g. `-XX:+UseZGC`

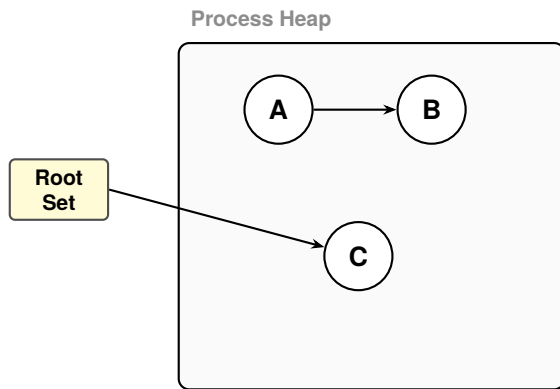


Garbage Collection



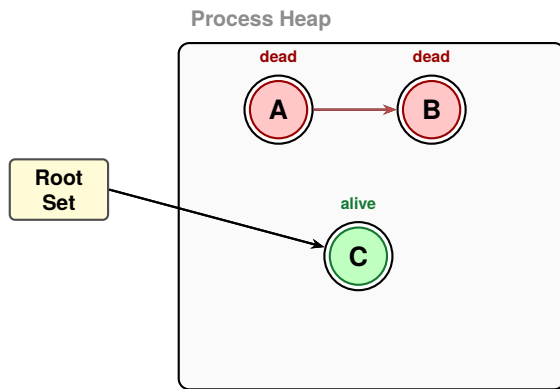
- Objects live on the **heap**; the root set is always live

Garbage Collection



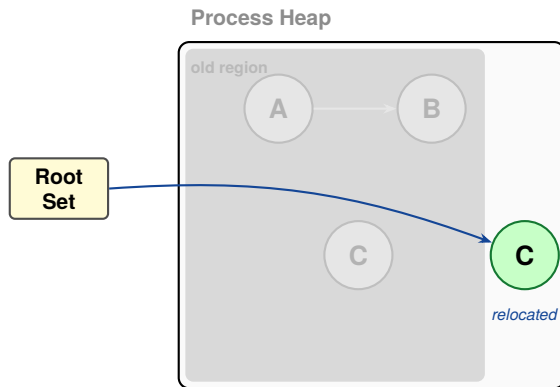
- ▶ Objects live on the **heap**; the root set is always live
- ▶ GC **marks**: follows all references reachable from roots

Garbage Collection



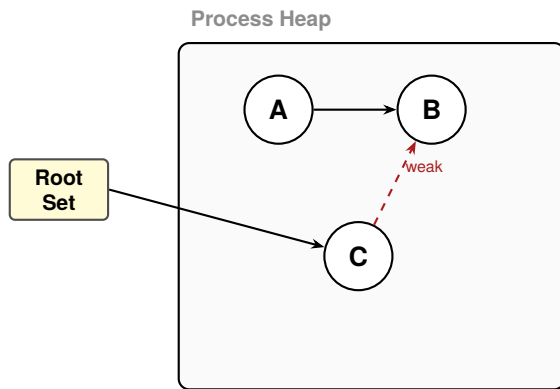
- ▶ Objects live on the **heap**; the root set is always live
- ▶ GC **marks**: follows all references reachable from roots
- ▶ **A and B** are unreachable – **dead** (even though $A \rightarrow B$); **C** is reachable – **alive**

Garbage Collection



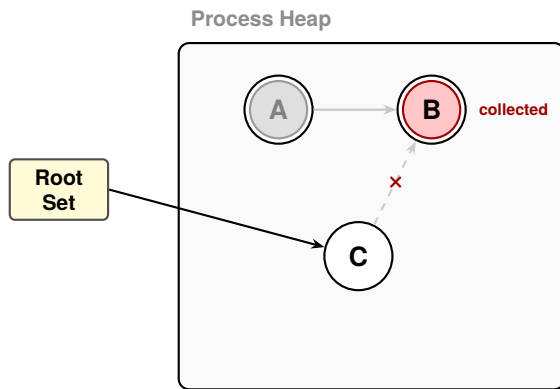
- ▶ Objects live on the **heap**; the root set is always live
- ▶ GC **marks**: follows all references reachable from roots
- ▶ **A** and **B** are unreachable – **dead** (even though $A \rightarrow B$); **C** is reachable – **alive**
- ▶ GC **relocates** C to a new region; old memory freed

Weak References



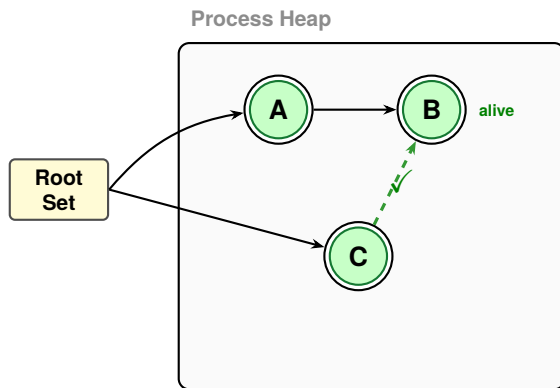
- ▶ A **weak reference** does not keep an object alive
- ▶ It does not count as reachability for GC purposes

Weak References



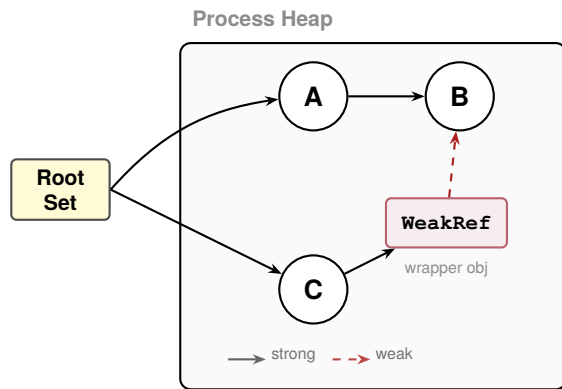
- ▶ A **weak reference** does not keep an object alive
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- ▶ If B has no *strong* path from roots: GC collects B; weak ref becomes `null`

Weak References



- ▶ A **weak reference** does not keep an object alive
- ▶ It does not count as reachability for GC purposes
- ▶ If B has no *strong* path from roots: GC collects B; weak ref becomes `null`
- ▶ Add strong ref root $\rightarrow$ A: B reachable through A
The weak ref from C to B is **valid**

The WeakReference Wrapper Object

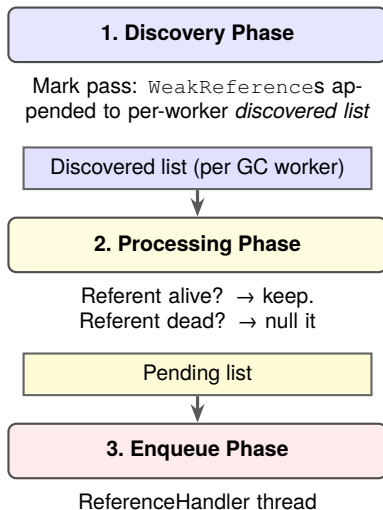


- ▶ The wrapper object exists to support the `ReferenceQueue` callback
- ▶ When the GC collects B, the `WeakReference` is enqueued so the application can react

```
// No queue
new WeakReference<>(obj)

// With notification queue
new WeakReference<>(obj, queue)
```

Reference Processing Pipeline



- ▶ During marking: found `WeakReferences` go onto a per-worker linked list
- ▶ After marking: check liveness per ref, clear dead ones
- ▶ Cleared refs handed to the `ReferenceHandler` thread for queue delivery
- ▶ Queue-less refs travel through all three phases – silently dropped at step 3

Motivation

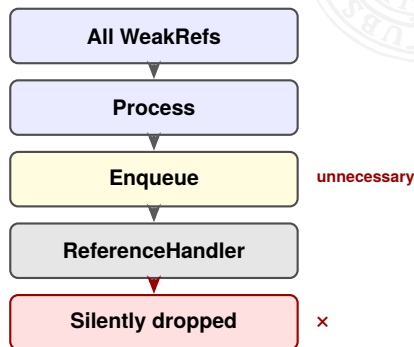
The Problem with Queue-less Weak References

- ▶ GC performance directly impacts application performance
- ▶ ZGC was designed for low latency – every unnecessary operation matters
- ▶ A JDK bug report (JDK-8029205) identified a concrete issue:

The issue

When a `WeakReference` *without* a `ReferenceQueue` is cleared, it is still handed to the `ReferenceHandler` thread – which then silently drops it, since there is no queue.

Unnecessary work for every single queue-less weak reference.



Research Questions

RQ1 – Pipeline Optimisations

How do specific modifications to ZGC's processing of `WeakReference` objects *without* a `ReferenceQueue` affect reference-processing time, total GC cycle time, GC memory footprint, and Java heap usage?

RQ2 – Weak Fields

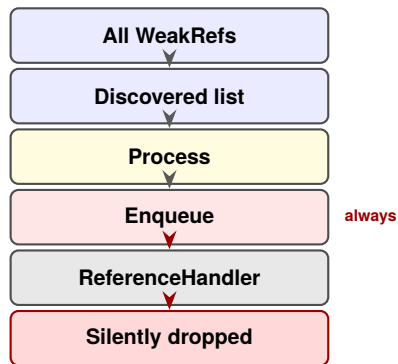
How does expressing weak semantics through *annotated object fields* affect the same metrics, compared to the optimised `WeakReference` pipeline approach?

RQ1: optimise the *existing* pipeline without changing how weak refs are represented

RQ2: eliminate the `WeakReference` wrapper object entirely

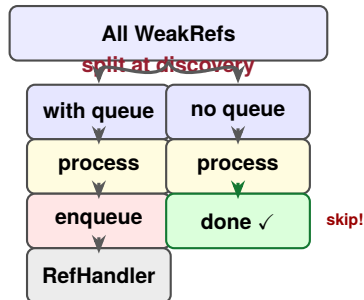
Design & Implementation

Optimisation 1: Skip Enqueue Path



- ▶ Every weak ref goes through the full pipeline
- ▶ Queue-less refs still hit the `ReferenceHandler` – which drops them
- ▶ Pure waste for the majority of weak references

Optimisation 1: Skip Enqueue Path

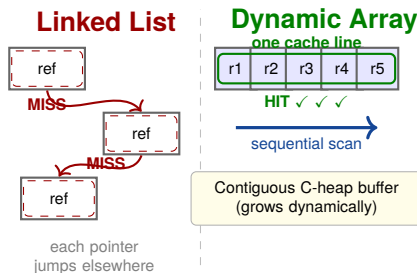


- ▶ At discovery: check the `ReferenceQueue` field
- ▶ Route queue-less refs to a *separate* discovered list
- ▶ No-queue pipeline: process (check + clear) – then **stop**
- ▶ With-queue: full existing behaviour

Why two lists instead of a branch?

A branch requires a memory load per ref to check the queue field. Two specialised loops avoid that per-reference cost.

Optimisation 2: Dynamic Array Discovered List

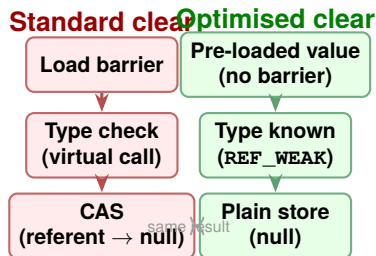


- ▶ Linked list: every next-node pointer is a **cache miss**
- ▶ Nodes scattered across memory – poor spatial locality
- ▶ Array: one cache load fills several consecutive entries
- ▶ Dynamic: grows amortised $O(1)$ per append
- ▶ Pre-loads referent address at discovery (used by Opt 3)

Trade-off

Higher auxiliary GC memory for

Optimisation 3: Optimised Clear Path



- ▶ CAS guards against concurrent application writes
- ▶ **Key observation:** the only application-visible write to a `WeakReference`'s referent field is `WeakReference.clear()` – which always writes `null`
- ▶ If a race occurs, both sides write `null` – same outcome regardless of order
- ▶ $\Rightarrow$ The CAS is unnecessary; a plain store is safe

Why only old-gen queue-less refs?

Research Question 2: Weak Fields (@weak)

WeakReference variant



@weak field variant



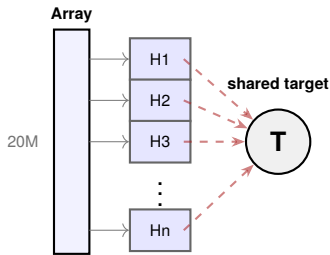
- ▶ A field annotated `@weak` is treated as a weak reference by ZGC – no wrapper object needed
- ▶ Discovered during marking, stored in a dynamic array of weak fields
- ▶ Cleared the same way – but **cannot** use the optimised clear path (a regular field can be written with any value)

```
// Before
class Holder {
    WeakReference<T> ref;
}
// After
class Holder {
    @weak T ref;
}
```

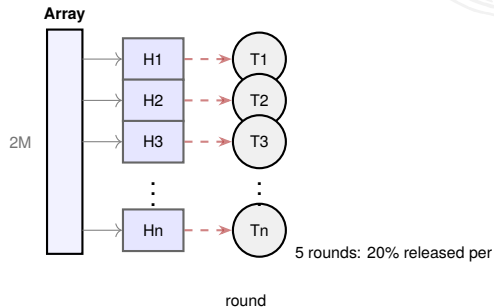
Evaluation

Benchmark Design

Single-Object Benchmark



Multi-Object Benchmark



- ▶ **Single:** 20M holders, one shared payload – all 20M refs processed in one GC cycle
- ▶ **Multi:** 2M holders, individual payloads, 5 gradual rounds – more realistic
- ▶ **Both:** no `ReferenceQueue`; both have a `@weak` field counterpart

Variants and Execution

Variant	clr	sep	dyn
none	—	—	—
dyn_only	—	—	✓
sep_only	—	✓	—
sep_dyn	—	✓	✓
clear_path_only	✓	—	—
clear_path_dyn	✓	—	✓
clear_path_sep	✓	✓	—
all	✓	✓	✓
weak_fields	—	—	—

- All $2^3 = 8$ pipeline combinations plus `weak_fields`

Execution Environment

UPPMAX supercomputer (Pelle)

AMD EPYC 9454P (Zen 4)

48 physical cores, 768 GiB RAM

4 parallel instances per run

Each: 10 JVM cores + 2 aux cores

100 GB heap `-XX:+UseZGC`

250 iterations (1 warmup discarded)

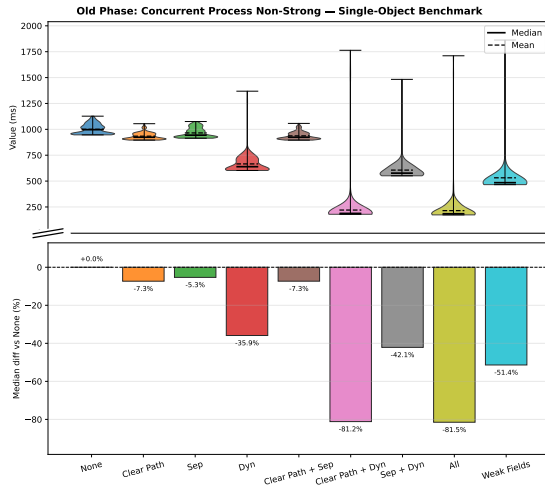
Reporting

Median over 250 iterations.

Violin + median percentage-difference histogram.

Results

Non-Strong Processing Time



Variant

Reduction

sep_only	-5%
clear_path_only	-7%
dyn_only	-36%
clear_path_dyn	-81%
all	-81%
weak_fields	-51%

Super-additive interaction

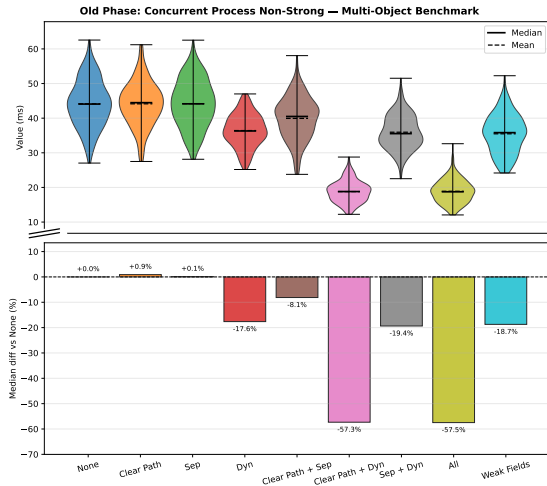
clear_path alone: -7%

dyn alone: -36%

Together: -81%

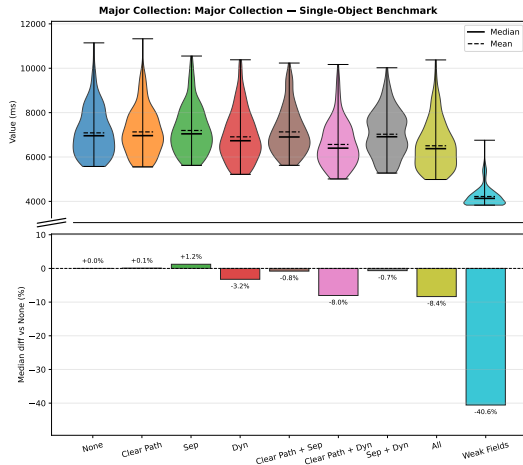
Each removes a different bottleneck.

Non-Strong Processing – Multi-Object & weak_fields

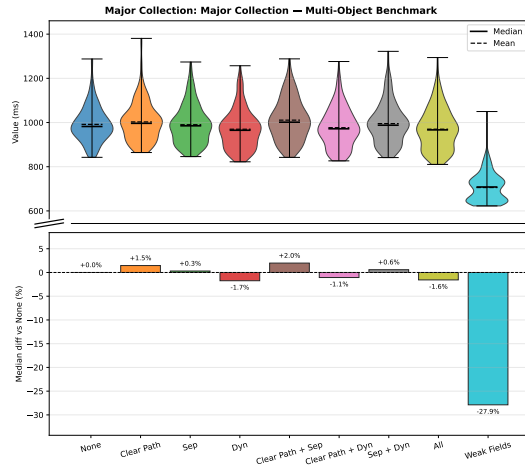


- ▶ Multi-object: same pattern – `clear_path_dyn` and `all` at -57%
- ▶ `sep_only`: 0% – the ReferenceHandler overhead is not captured by this metric
- ▶ `weak_fields`: -51% (single), -19% (multi)
- ▶ Comparable to `sep_dyn` – because `@weak fields` cannot use the optimised clear path

Major Collection Time

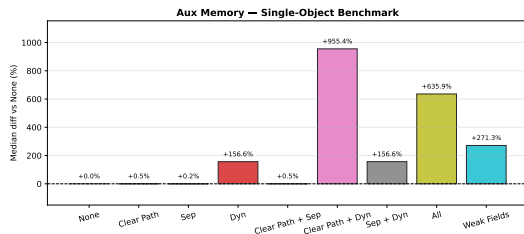


Single-object benchmark



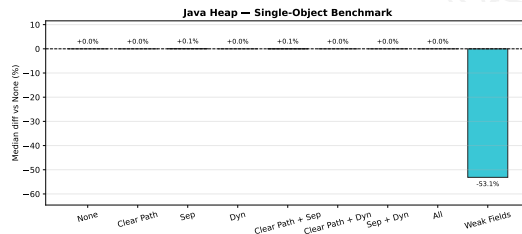
Multi-object benchmark

Memory Usage



Auxiliary GC memory – single-object benchmark

- ▶ `clear_path_dyn`: ×10 aux memory (1268 MB vs 120 MB)
- ▶ `all`: ×7 (sep reduces bytes per entry vs dyn alone)



Java heap – single-object benchmark

- ▶ **All** WeakReference variants: heap essentially identical
- ▶ `weak_fields`: **−53%** heap (single), **−5%** (multi) – no wrapper objects to allocate and promote

Conclusions

Conclusions

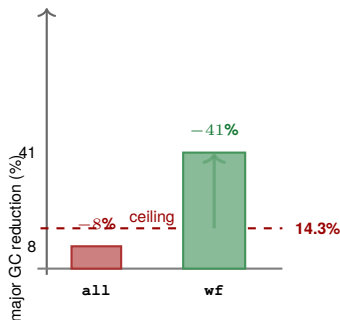
RQ1 – Pipeline Optimisations

- ▶ **81%** reduction in non-strong processing time (`clear_path_dyn / all`)
- ▶ Translates to only **8%** in total major collection time
- ▶ Hard ceiling: non-strong is 14.3% of major GC time (single) and 4.5% (multi)

RQ2 – Weak Fields

- ▶ **41%** major GC time reduction (single), **28%** (multi)
- ▶ **53%** heap reduction (single) – no wrapper objects
- ▶ Structural advantage: fewer objects means less work across *every* GC phase

Key Takeaway



Optimise the pipeline (`all`)

–81% non-strong, but only –8% total GC

Hard ceiling: non-strong is only 14% of major GC

Change the representation (`weak_fields`)

–41% total GC, –53% Java heap

Fewer objects → less work in every GC phase

The bottleneck is the *representation*

Rethink how weak semantics is represented –
not just how to process the wrappers faster.

Thank You

Questions?

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